

DIHWorld Audio

Bass / Low End Technical Manual

Current product versions, controls, presets, meters, and placement

DIHWorld Audio builds connected production systems around the Imhotep 369 Engine—a Triadic Harmonic Lattice—and Ma'at 357 / MomentumBend. Phi-derived control and the canonical Phi/2 ceiling preserve movement, balance, and musical intent without brickwall limiting.

This edition documents 4 current products. Release numbers are taken from the installed DIHWorld Audio Audio Units verified on 2026-09-05; they are product versions, not a shared catalog version.

Canon reference: normalized $\Phi/2 = 0.809016994$. MomentumBend shapes motion and containment; the ceiling is a canonical boundary, not permission for hard limiting.

Operating law: begin from reset or a purpose-matched preset, compare at equal loudness, change one system at a time, and judge the result in the full arrangement.

MARIANA

RELEASE	FAMILY	PLACEMENT
v1.0.1	Bass / Low End	Low End Source

Sub and foundation processor with depth, sub pressure, kick definition, target mode, and parallel mix. A named optional Sidechain drives kick-aware reduction only when Sidechain Key is enabled, leaving the audible main path neutral otherwise.

Recommended use: low-end/sub field

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Bool
depth	Depth	Float
pressure	Sub Pressure	Float
foundation	Foundation	Float
kick_define	Kick Define	Float
mode	Target Mode	Choice
input	Input	Float
mix	Mix	Float
gainmatch	Gain Match	Bool

Factory starting points

MARIANA SWEET SPOT • LOW GRAVITY CLEAN • WARM SUBFIELD • SUB WEIGHT • DEEP PRESSURE • ABYSSAL FIELD • TRENCH FLOOR • GENTLE FINISH • OCEAN FLOOR • SUB ENHANCED • KICK PRESSURE • ABYSS BLOOM • TIDAL TRACE • SHALLOW CURRENT

MONOLITH

RELEASE	FAMILY	PLACEMENT
v1.0.5	Bass / Low End	Low End Source

Low-end impact and body shaper with excavation, weight, resonance, reverse motion, drive, and balance. A named optional Sidechain supports kick-controlled Stone Duck and remains inactive until Sidechain Key is enabled.

Recommended use: low-end mass

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
gate	gate	Referenced
input	input	Referenced
output	output	Referenced
channelmode	channelmode	Referenced
gainmatch	gainmatch	Referenced
excavation	excavation	Referenced
impact	impact	Referenced
weight	weight	Referenced
resonance	resonance	Referenced
reverse	reverse	Referenced
drive	drive	Referenced
balance	balance	Referenced

Factory starting points

MONOLITH PHI IMPACT [Stereo] • MONOLITH PHI IMPACT [Mono] • MONOLITH TRANSIENT STONE [Stereo] • MONOLITH TRANSIENT STONE [Mono] • MONOLITH BODY WEIGHT [Stereo] • MONOLITH BODY WEIGHT [Mono] • MONOLITH SHAPE IMPACT [Stereo] • MONOLITH SHAPE IMPACT [Mono] • MONOLITH BUS CARVE [Stereo] • MONOLITH BUS CARVE [Mono] • MONOLITH GATEWAY [Stereo] • MONOLITH GATEWAY [Mono] • MONOLITH PHI IMPACT [Dual Mono] • MONOLITH TRANSIENT STONE [Dual Mono] • MONOLITH BODY WEIGHT [Dual Mono] • MONOLITH SHAPE IMPACT [Dual Mono] • MONOLITH BUS CARVE [Dual Mono] • MONOLITH GATEWAY [Dual Mono]

Meters and visualizers

INPUT / OUTPUT: Live pre/post level. Use for bypass and level matching.

IMPACT: Low-end transient activity. Shows the punch component being produced.

WEIGHT: Sustained low-frequency body. Shows foundation rather than peak attack.

RESONANCE: Active resonant low-end energy. Persistent high readings warn of ringing.

REVERSE: Reverse-motion contribution. Shows how strongly the reverse path is active.

Monolith field: Combined low-end activity map. Reactive summary, not a spectrum analyzer.

DEPTH_BOOST_PRO

RELEASE	FAMILY	PLACEMENT
v1.0.1	Bass / Low End	Low End Source

Focused low-end depth extension built on delay-based mid/side field placement. Its Spatial Gain control (0 dB down to -12 dB) is applied after the depth field, making DEPTH_BOOST_PRO a spatial gain-reduction stage as well as a depth/width processor. A high-value hybrid is DEBO -> DEPTH_BOOST_PRO: drive DEBO for pressure and density, then reduce Spatial Gain to return the energized result to fair level while retaining the chosen field.

Recommended use: low/depth support and post-field spatial gain reduction via dedicated Spatial Gain (0 dB to -12 dB); blue secondary dot

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
input	Input Trim	Float
gainmatch	Gain Match	Bool
depth	Spatial Depth	Float
output	Spatial Gain	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Factory starting points

PHI ROOT TOUCH [Stereo] • TIGHT KICK ROOT [Stereo] • VOCAL SHADOW LIFT [Stereo] • PHI SPACE OPEN [Stereo] • LOW END GLUE [Stereo] • SUB FLOOR WEIGHT [Stereo] • BASS WIDTH SPREAD [Stereo] • MIX FRONT TO BACK [Stereo] • DRUM ROOM DEPTH [Stereo] • TEMPLE WIDTH [Stereo] • CINEMA FLOOR [Stereo] • LOW FIELD EXTENDER [Stereo] • MONO BASS FOCUS [Stereo] • FULL BUS DEPTH [Stereo] • PHI ROOT TOUCH [Mono] • TIGHT KICK ROOT [Mono] • VOCAL SHADOW LIFT [Mono] • PHI SPACE OPEN [Mono] • LOW END GLUE [Mono] • SUB FLOOR WEIGHT [Mono] • BASS WIDTH SPREAD [Mono] • MIX FRONT TO BACK [Mono] • DRUM ROOM DEPTH [Mono] • TEMPLE WIDTH [Mono] • CINEMA FLOOR [Mono] • LOW FIELD EXTENDER [Mono] • MONO BASS FOCUS [Mono] • FULL BUS DEPTH [Mono] • PHI ROOT TOUCH [Dual Mono] • TIGHT KICK ROOT [Dual Mono] • VOCAL SHADOW LIFT [Dual Mono] • PHI SPACE OPEN [Dual Mono] • LOW END GLUE [Dual Mono] • SUB FLOOR WEIGHT [Dual Mono] • BASS WIDTH SPREAD [Dual Mono] • MIX FRONT TO BACK [Dual Mono] • DRUM ROOM DEPTH [Dual Mono] • TEMPLE WIDTH [Dual Mono] • CINEMA FLOOR [Dual Mono] • LOW FIELD EXTENDER [Dual Mono] • MONO BASS FOCUS [Dual Mono] • FULL BUS DEPTH [Dual Mono]

Meters and visualizers

Touch sphere: Depth field plus live level. Water motion follows field/level activity; sphere position remains the output control.

Spatial field: Depth, field energy, and motion. A reactive placement view, not a calibrated spectrum.

NTRPL

RELEASE	FAMILY	PLACEMENT
v1.0.2	Bass / Low End	Low End Source

Low-end transformation engine for rewrite map, novelty, interval motion, contour, shape, space, and originality guard. A named optional Sidechain supports keyed Rewrite Duck and remains inactive until Sidechain Key is enabled.

Recommended use: reclassified bass family

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
chordMap	Rewrite Map	Choice
interpolate	Novelty	Float
motif	Chord Rewrite	Float
rhythm	Motion	Float
pitch	Interval Distance	Float
melody	Contour	Float
shape	Shape	Float
field	Space	Float
guard	Originality Guard	Float
alternateMix	Mix	Float
output	Output	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Factory starting points

01 SOURCE TRACE [Stereo] • 02 PHI BODY REWRITE [Stereo] • 03 LOW WEIGHT LATTICE [Stereo] • 04 HARMONIC REWRITE [Stereo] • 05 ALT PHI FIELD [Stereo] • 06 WIDE RECONTOUR [Stereo] • 07 DEEP BASS BLOOM [Stereo] • 08 EDGE TRANSLATION [Stereo] • 01 SOURCE TRACE [Mono] • 02 PHI BODY REWRITE [Mono] • 03 LOW WEIGHT LATTICE [Mono] • 04 HARMONIC REWRITE [Mono] • 05 ALT PHI FIELD [Mono] • 06 WIDE RECONTOUR [Mono] • 07 DEEP BASS BLOOM [Mono] • 08 EDGE TRANSLATION [Mono] • 01 SOURCE TRACE [Dual Mono] • 02 PHI BODY REWRITE [Dual Mono] • 03 LOW WEIGHT LATTICE [Dual Mono] • 04 HARMONIC REWRITE [Dual Mono] • 05 ALT PHI FIELD [Dual Mono] • 06 WIDE RECONTOUR [Dual Mono] • 07 DEEP BASS BLOOM [Dual Mono] • 08 EDGE TRANSLATION [Dual Mono]

Meters and visualizers

- SOURCE:** Original input contribution. Shows how much source survives into the main route.
- REWRITE:** Generated harmonic-rewrite activity. Shows the newly synthesized low-end material.
- ALT OUT:** Alternative new-signal output. Always represents the generated signal path.
- MAIN:** Final blended output. Use against SOURCE for level and blend context.
- Harmonic Rewriter:** Live input-to-output contour. Shows source energy changing into rewritten energy.
- Rewrite Map:** Novelty, chord, interval, contour, motion, space, guard, and mix state. A live process map; it is not a frequency analyzer.